

181551/121551/073651

a) Light intensity control

Q.28 What is chopper & define all types of choppers.

Q.29 Write a note on cycloconverter.

Q.30 Discuss about inverters explain series & parallel inverters.

Q.31 What is UPS & explain the working of Online & off line UPS.

Q.32 Explain the working of series inverters.

Q.33 Discuss about SMPS.

Q.34 Write a short note on DIAC.

Q.35 Discuss about the different method of SCR triggering.

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain construction & working principles of UJT with its characteristics.

Q.37 Write a short notes on any two:-

a) Variable Freq. Drive

b) HVDC

c) Battery Charge using Thyristor

Q.38 Describe single phase bridge type cycloconverter.

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5th Sem / IC, EI
Subject:- Power Electronics

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 An SCR is a :-

a) Two layer two junction device

b) Four layer three junction device

c) Three layer two junction device

d) Four layer four junction device

Q.2 That no. of gates in a silicon controlled switch is:-

a) One

b) Two

c) Three

d) Four

Q.3 Adiac can be fired:-

a) Only by negative half cycle of supply

b) Only by positive half cycle of supply

c) Both by positive as well as negative half cycle of supply

d) Without any biasing voltage.

Q.4 An SCR can be prepared by sand witching:-

a) Two NPN transistor

b) Two PNP transistor

c) One NPN & one PNP transistor

d) Any one of these all options

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- Q.5 A single phase full convert can operate in
 a) 4 quadrants b) 3 quadrants
 c) 2 quadrants d) 1 quadrants
- Q.6 A semi converter rectifier contains:-
 a) Diodes
 b) Thyristors
 c) Mixture of Diodes Thyristors
 d) Diode & Freewheeling diode
- Q.7 An inverter converts:-
 a) DC into variable DC
 b) DC into AC
 c) AC into DC
 d) AC into AC of different frequency
- Q.8 The duty cycle of chopper is given by:-
 a) $\frac{T_{ON}}{T_{off}}$ b) $\frac{T_{ON}}{T_{ON}}$
 c) $\frac{T_{ON}}{T_{on} + T_{off}}$ d) $\frac{T_{off}}{T_{on} + T_{off}}$
- Q.9 Cycloconverter drives are generally employed in:-
 a) Traction
 b) Milling
 c) Generating low frequency
 d) Generating Pulse
- Q.10 UPS is never used in
 a) Street Lighting
 b) Communication links
 c) Computer
 d) All of above

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 An SCR is more efficient than a triac. (True/False)
- Q.12 Draw the symbol of SCR.
- Q.13 Why a heat sink required with a Thyristor.
- Q.14 SCR is the oldest & widely used member of _____ family.
- Q.15 A semi-converters can operate in all four quadrant. (True/False)
- Q.16 What is firing angle?
- Q.17 A cycloconverter is a device which convert _____
- Q.18 What are the applications of Cycloconverters.
- Q.19 Name the two methods of speed control of DC motors.
- Q.20 Define UPS.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Discuss about construction & working of UJT & also draw the diagram also.
- Q.22 Define series operation of SCR.
- Q.23 Discuss about construction & working of SCR & also draw the diagram also.
- Q.24 Draw V-I characteristics of SCR with explanation.
- Q.25 Draw V-I characteristics of TRIAC.

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